

SHREEJIT VERMA

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PROFESSIONAL EXPERIENCE

BNP Paribas CIB, New York, USA

C++ Quantitative Developer (Co-op), Automated Market Making

Feb 2026 – May 2026

- Built low-latency components of the automated market-making stack for the Prime Credit Market (average **\$500M** of daily market-making volume), spanning real-time market-data ingestion, tick analytics, and pricing/execution paths
- Profiled and optimized the software hot path feeding FPGA-accelerated market-data handlers and quoting engines
- Integrated secure on-premise LLM tooling with Git/Jira/Confluence to automate code, testing, and documentation workflows

LogiNext Solutions Inc., Mumbai, India

Senior Software Engineer, Analytics

Mar 2023 – Jul 2024

- Architected and developed Map Construction Algorithms, Map Routing Algorithms, and solved Rich Vehicle Routing Problems (3 Nested NP-Hard Problems) using CP-SAT Constraint Programming and Convex optimization over PostGIS, MongoDB and S3
- Led a 12-engineer team delivering a high-throughput geospatial mapping application platform
- Built an LLM-powered debugging and query-resolution tool across the company, cutting mean bug-resolution time by **80%**

Versor Investments (QR Systems LLP), Mumbai, India

Quantitative Developer, Merger Arbitrage and Stock Selection Portfolio

Feb 2022 – Oct 2022

- Developed and backtested systematic merger-arbitrage strategies of an **\$8.5 Billion** AUM fund, improving alpha capture by **15%**
- Built and deployed ML pipelines for Order and Execution Management Systems, increasing trade execution efficiency by **29%**
- Designed an ESG-driven merger-arbitrage signal subsequently embedded across the firm's existing portfolios

Bank of America (BA Continuum India Pvt Ltd.), Chennai, India

Senior Software Engineer, Fixed Income Commodities and Currencies (FICC)

Jan 2020 – Jul 2021

- Engineered Python trading services for trade processing, storage, matching, and execution on QUARTZ platform
- Built C++ pipelines to store trades into the SANDRA object-oriented database, reducing trade processing latency by **50%**
- Led the migration of **1 million** plus lines of code to Python 3.8, improving scalability and execution efficiency by **40%**

Senior Tech Associate, Data Analysis and Insight Technology

Jun 2018 – Dec 2019

- Architected and developed an ML/AI platform to deploy predictive models, increasing decision-making accuracy by **67%**
- Designed machine learning models for data validation rules prediction, equivalent to **36 Full-Time Equivalents (FTEs)**

RESEARCH & SELECTED PROJECTS

AI-Integrated FPGA for Market Making in Volatile Environments (MS Thesis, Stevens) | [GitHub](#)

Oct 2025 – May 2026

- Engineered a sub-10 μ s trading platform with a custom-built limit order book, FPGA market data handlers, DPDK kernel bypass, hardware timestamping, and lock-free queues for deterministic microsecond-scale execution

Adaptive Volatility Regime Based Execution and Risk Framework | [GitHub](#)

Sep 2025 – Dec 2025

- Developed adaptive volatility regime switch framework dynamically selecting among passive, TWAP, and aggressive strategies
- Achieved **20.0%** increase in Sharpe Ratio, **6.1%** transaction cost reduction, **20.1%** CVaR decrease with robust risk management

Statistical Arbitrage Reversal and Momentum Strategies (WallStreetQuants) | [GitHub](#)

Jun 2025 – Aug 2025

- Designed and backtested a 120-day volume-momentum-based crypto portfolio strategy, yielding a **155.76%** annualized return and **1.94** Sharpe Ratio net of transaction costs, significantly outperforming the Bitcoin buy-and-hold benchmark

Dynamic Portfolio Optimization (MScFE Thesis, WorldQuant University) | [GitHub](#)

Mar 2024 – Jun 2024

- Built a real-time portfolio optimizer using convex and non-convex methods with adaptive rebalancing and multi-factor risk modeling, enhancing risk-adjusted returns across rate, FX, credit, and market risks

SKILLS

Mathematics/Statistics: Probability, Stochastic Calculus, PDEs, Linear Algebra, Numerical Methods, Markov Processes

Quant/ML: Statistical Analysis, Derivative Pricing, Time Series Analysis, Factor Modeling, Predictive Modeling, Greeks, NumPy, Pandas, Polars, SciPy, PyTorch, TensorFlow, Scikit-learn, RAG, QuantLib, Statsmodels, QtPy, CVXPY, OpenMP, MPI, CUDA

Machine Learning: Linear Regression, Clustering, Random Forest, XGBoost, RNN, LSTM, Deep Learning, Neural Networks, NLP

Programming: C++ (17/20/23, primary), Python, C, Java, R, MATLAB, JavaScript, Node.js, ReactJS, Bash Scripting

Data Engineering: Spark, PySpark, Airflow, Dask, FastAPI, REST APIs, Flink, NoSQL, ZeroMQ, PostgreSQL, SQL, KDB+, Q

Low latency: TCP/IP, UDP, Multicast, DPDK/kernel bypass, FPGA (Verilog, VHDL),

Cloud Computing and DevOps: Linux, Git, CI/CD, Jenkins, Ansible, Docker, Kubernetes, Helm, AWS, GCP, WSL

EDUCATION

Georgia Institute of Technology – MS, Computer Science (Computing Systems)

Aug 2024 – Dec 2026 (exp.)

Stevens Institute of Technology – MS, Financial Engineering | **GPA:** 3.974/4.0

Aug 2024 – May 2026

WorldQuant University – MS, Financial Engineering | **GPA:** 86%

Dec 2021 – May 2024

Vellore Institute of Technology – B.Tech, Computer Science and Engineering | **GPA:** 8.78/10.0

Jul 2014 – Sep 2018

HONORS & INTERESTS

CFA Level 1 (passed) | Bloomberg Market Certification | Global Recognition Gold Award (Bank of America) | Beta Gamma Sigma

1st Place (personal portfolio), Vanguard ETF Trading Challenge | President of Stevens Graduate Financial Association

Interests: Chess, Poker, F1, Martial Arts, Cricket, Boxing, Badminton, Reading, Cooking, Dancing, Psychology, History, Philosophy